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2.4.9. Class Aves

(L. *Avis* –bird)

Aves are commonly known as birds. The characteristic feature of Aves is the presence of feathers and the ability to fly except for flightless birds (Eg. *Ostrich, Kiwi, Penguin*). The forelimbs are modified into wings, and the hind limbs are adapted for walking, running, swimming and perching. The skin is dry and devoid of glands except the oil gland or preen gland at the base of the tail. The exoskeleton consists of epidermal feathers, scales, claws on legs and the horny covering on the beak. The endoskeleton is fully ossified (bony) and the long bones are hollow with air cavities (pneumatic bones). The pectoral muscles of flight (pectoralis major and pectoralis minor) are well developed. Respiration is by compact, elastic, spongy lungs that are continuous with air sacs to supplement respiration. The heart is four chambered. Aves are homeothermic. Migration and parental care is well marked. Urinary bladder is absent. Sexes are separate with well marked sexual dimorphism. In males, the testes are paired but in females, only the left ovary is well developed while the right ovary is atrophied. All birds are oviparous. Eggs are megalecithal and cleidoic. Fertilization is internal.

Examples *Corvus* (Crow), *Columba* (Pigeon), *Psittacula* (Parrot), *Pavo* (Peacock), *Aptenodytes* (Penguin), *Neophron* (Vulture), *Chalcophaps indica* (Tamilnadu state bird, Common Emerald Dove) (Figure 2.28).



Hooded Pitohui

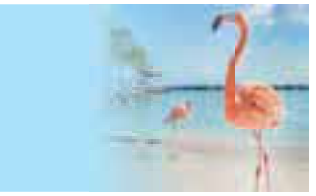
(*Pitohui dichrous)*

The Hooded Pitohui is a songbird found in the rain forests of New Guinea, The first poisonous bird to be documented A neurotoxin called Homobatrachotoxin is found in its skin and feathers, causes numbness and tingling in those touching the bird.





Vulture



Flamingo



Common Emerald Dove (Tamil Nadu State Bird)



Woodpecker



Penguin



Ostrich



Kiwi



Humming Bird

Figure 2.28 Examples of Aves